ALEXANDRU RUSU

519-591-0245 a3rusu@uwaterloo.ca LinkedIn GitHub

EDUCATION

University of Waterloo Cumulative GPA: 3.95
BASc in Mechatronics Engineering

- **Varsity Swimming:** Currently competing for the university swim team; competed at the 2025 National Trials.

SKILLS

- **Programming:** Python, C++, Embedded C++, C#, JavaScript, TypeScript, HTML, CSS, React, Node.js, Astro
- **Embedded & Hardware:** ESP32, Arduino, Custom PCBs, Sensors (current, distance, colour), IoT, Serial Comms
- **Tools & Software:** Autodesk Inventor, KiCad, CAD/CAM, GD&T, MediaPipe, librosa, Machine Vision, MATLAB
- **Development:** Git, GitHub, GitLab, REST APIs, HTTP Requests, Object-Oriented Programming, API Design

EXPERIENCE

Manufacturing Process Engineering Co-op *Hammond Manufacturing* Apr 2026 - Present

- Applied **Design for Manufacturing (DFM)** to evaluate enclosure cutouts and maximize production output.
- Managed comprehensive **CAD/CAM** workflows to hold custom part tolerances within **10 thou** of specification.
- Collaborated with cross-functional manufacturing teams to troubleshoot line issues and maintain part quality.
- Caught defective parts through in-process quality checks, saving an estimated **$1,000s** in scrap and rework.
- Streamlined daily process documentation and assembly testing procedures, saving hours of production weekly.

Embedded Systems Intern *NeuronicWorks* Jul 2024 - Aug 2024

- Developed an embedded monitoring system using an AC current draw **sensor** to infer coffee machine brewing state.
- Programmed **C++ firmware** for an IoT **ESP32** Wi-Fi module sending **HTTP requests** to backend servers.
- Built a **Mattermost bot** alongside the engineering team to deliver automated coffee machine-state notifications.
- Executed **20+** comprehensive calibration tests under real operating conditions, achieving ~**95%** detection accuracy.
- Collaborated with hardware engineers during early prototyping, gaining exposure to device build workflows.
- Presented the final **IoT** prototype and calibration data to senior leadership, highlighting performance metrics.

Web Developer *Levanta Labs* Feb 2025 - Jun 2025

- Developed and deployed production-ready web pages using the **Astro framework** and static site generation.
- Built 10+ responsive front-end components using **TypeScript** and **HTML/CSS** for adaptive cross-device styling.
- Implemented a **Slack webhook** integration routing user contact details to the team, accelerating lead response.
- Collaborated with senior developers to maintain component consistency and brand adherence across all pages.
- Conducted thorough cross-browser testing protocols to ensure seamless performance across supported devices.

PROJECTS

Automated Metrology Setup *In Progress*

- Building a custom **machine vision** inspection station using an **ESP32**, **custom PCBs**, and active sensor arrays.
- Developing software overlaying part cutout tolerances on live camera feeds for **precise** in-line enclosure inspection.
- Designed a specialized physical ring light rig to optimize image contrast and improve **machine vision reliability**.
- Calibrated active **distance sensors** for consistent measurement accuracy across diverse electronic enclosures.

Blinkrite (In Progress) Blinkrite

- Designing a smart monitor lamp that tracks user blink rates via **MediaPipe** to trigger adaptive bias lighting.
- Developed an **ML pipeline** processing real-time webcam feeds with dual detection models at 15 and 30 fps.
- Engineered custom physical hardware with an **ESP32** receiving serial/network commands to modulate light output.
- Designed and manufactured a custom **PCB** and mount to integrate the complete system onto any user monitor.

2D Plotter Robot Plotter Project

- Designed and built a **2-axis gantry** plotter robot autonomously drawing shapes via real-time sensor integration.
- Implemented a custom **rack-and-pinion motion system** using 3D-printed components and off-the-shelf hardware.
- Programmed the robot in **C++** to continuously read sensors and dynamically control physical motor outputs.
- Implemented autonomous colour-triggered drawing and a manual free-draw mode via a handheld remote control.

Beatdown (In Progress) Beatdown

- Developing a multiplayer guessing game with a **real-time database** synchronizing game state across all clients.
- Programmed a scalable single-player **AI opponent** with adjustable difficulty levels to simulate real player logic.
- Designed an intuitive user interface and interactive game loop for smooth, responsive real-time network gameplay.